LESSON 104 (1.0 GROUND and 1.0 AATD)

Lesson Objective

During this lesson, the student will gain a better understanding of previous material and become familiar with operation of the primary and secondary aircraft controls and their effect on the aircraft in steady state flight. The student will learn how to select, trim and hold an attitude as well as the effect and operation of ancillary controls.

Briefing

- Outline the objectives and completion standards of the lesson.
- Establish the student and instructor responsibilities.
- Review homework assigned in the previous lesson.
    - Quiz student on checklist memory items as needed.
    - Review the prepared TOLD card for the lesson.
    - Quiz the student on the readings as needed.
        - Read chapter 03 of the Airplane Flying Handbook - Basic Flight Maneuvers:
            - The four fundamentals.
            - Effects and use of flight controls.
            - Attitude flying.
            - Straight and level flight.
- Student questions.

References

- Advisory Circulars. (Search Tool)
    - Pilots' Role in Collision Avoidance. (AC 90-48)
- Pilot's Handbook of Aeronautical Knowledge. (FAA-H-8083-25B)
    - Chapter 06 - Flight Controls.
- Warrior III Pilot's Information Manual.
    - Section 06 - Weight and Balance.

Lesson Content (Ground)

Obtaining a Weather Briefing

- Quiz the student on:
    - How to satisfy the legal preflight weather briefing requirements?
    - Different ways to obtain a weather briefing.
    - Types of weather briefings.

Performance & Weight and Balance (TOLD card)

- Review and inspect the student's TOLD cards (as per the assigned homework from the previous lesson).
    - Quiz the student on:
        - Weight and balance (aft CG vs fwd CG).
        - Weather information.

- Pressure altitude and density altitude.
- Takeoff and landing distance.
- V-speeds.
- Go/no-go decision.

Effects of Flight Controls

- Primary:
    - Elevator (Stabilator) → Pitch.
    - Rudder → Yaw.
    - Ailerons → Roll.
- Secondary:
    - Elevator → Airspeed.
    - Rudder → Roll.
    - Ailerons → Yaw.
- The speed that air is moving over a flight control surface will greatly impact the effectiveness of said surface.
    - Faster airflow will result in a more responsive flight control.
    - Slower airflow will result in a less responsive flight control.
- This will be demonstrated in the simulator.

Operation of Systems - Primary and Secondary Flight Controls

- Refer to PA-28-161 System PowerPoint.
- Flight controls are actuated by the use of cables, pulleys, pushrods, and torque tubes.
- Primary flight controls:
    - Ailerons (differential):
        - Controlled with yoke moving left and right.
        - Airflow deflection - Newton's third law of motion.
        - Located on the outboard trailing edge.
        - Rolls on the longitudinal axis.
        - Counterweights – to help balance controls.
        - Move up 25 degrees and move down 12.5 degrees.
        - Helps compensate for adverse yaw.
            - The up aileron creates more parasite drag to compensate for the induced drag created by the downward aileron.
    - Stabilator:
        - Controlled with yoke moving forward and backward.
        - Combines horizontal stabilizer and elevator.
        - Trailing edge moves up 14 degrees and down 2 degrees.
        - Has an anti-servo tab on the trailing edge.
            - Moves in the same direction as stabilator.
            - Since the stabilator deflects so much airflow, the anti-servo tab increases the pressure felt on the yoke to avoid overcontrolling the airplane.

- Also acts as a trim tab.
  - In opposite directions.

- Rudder:
  - Controlled with rudder pedals.
  - Controls yaw on the vertical axis.
  - Vertical airfoil on trailing edge of vertical stabilizer.
  - Moves left and right 27 degrees (+/- 2).
  - Rudder trim below throttle quadrant.

- Secondary Flight Controls:
  - Flaps (slotted):
    - Operated with a lever, through a torque tube and push rods to the flaps.
    - Air flows through the gap between wing and flap. This air is able to re-energize the airflow on top of the wing. This gives the airplane the ability to maintain lift at slower airspeeds and hold a higher AOA.
    - An over center lock on the right flap works while up, to support weight.
    - 4 settings:
      - 0 degrees.
      - 10 degrees.
      - 25 degrees.
      - 40 degrees.
    - <u>Quiz the student on V_{FE} speed!</u>
  - Trim:
    - Used to decrease required pressure on flight controls.
    - Stabilator and rudder trim.
    - Stabilator trim:
      - Although the anti-servo tab moves in the same direction as the stabilator, whenever you trim, the tab doesn't move AS much.
    - <u>Emphasize the importance of trim!</u>

Visual Scanning and Collision Avoidance

- Emphasize that it is the pilot in command's responsibility to avoid traffic.
- FAR regulation 91.111 - operating near other aircraft.
  - Cannot operate near another aircraft to create a collision hazard.
  - Cannot fly in formation unless there was a prior agreement between the PIC's of participating aircraft.
  - No passengers for hire in formation flight.
- Explain how to scan for traffic in daytime conditions.
  - 10 degrees per second, looking both above and below aircraft level.
- Clearing procedures:
  - During taxi operations.
  - Before climbing.
  - During descents.

    - Clearing turns.
- IF YOU SEE SOMETHING, SAY SOMETHING!

Positive Aircraft Control

- Fly the aircraft first.
    - Aviate, navigate, communicate.
- Situational awareness.
- Positive exchange of flight controls.
    - Emphasize the importance of not releasing flight controls until the third statement in the exchange is made.

<u>Lesson Content (AATD)</u>

Checklist Procedures

- <u>Re-emphasize the importance of using checklists.</u>
- Have the student run through all of the checklists.
- Assist if needed.
- Introduce the “passenger briefing” and the “engine fire during start brief”.

Before Takeoff Check

- Have the student run through the checklist.
- Assist if needed.
- Introduce the “takeoff briefing”.

Visual Scanning and Collision Avoidance

- Emphasize that it is the pilot in command’s responsibility to avoid traffic.
- Demonstrate how to scan for traffic in daytime conditions.
    - 10 degrees per second, looking both above and below aircraft level.
- IF YOU SEE SOMETHING, SAY SOMETHING!

Power plant operation

- Introduce that we need to switch the fuel tank every 30 minutes!
- Engine is to be operated at 1000 RPM when stationary on the ground.
- Mixture should be left full rich below 3,000ft.

Primary Effects of Flight Controls

- <u>Emphasize to the student that all control movements must be smooth!</u>
- Elevator → Pitch.
- Rudder → Yaw.
- Ailerons → Roll.

Secondary Effects of flight Controls

- Elevator → Airspeed.
- Rudder → Roll.
- Ailerons → Yaw.
- Introduce the concept of a spiral dive and how to properly recover.

Effects of Airspeed

- The speed that air is moving over a flight control surface will greatly impact the effectiveness of said surface.
  - Faster airflow will result in a more responsive flight control.
  - Slower airflow will result in a less responsive flight control.

Effect of Slipstream

- Fast:
  - Elevator → More Pitch.
  - Rudder → More Yaw.
  - Ailerons → More Roll.
- Slow:
  - Elevator → Less Pitch.
  - Rudder → Less Yaw.
  - Ailerons → Less Roll.

Effect of Trim

- <u>Emphasize the importance of trim!</u>
- Natural sense.
  - Roll wheel forward for forward trim.
  - Roll wheel aft for aft trim.
- Coarse versus fine trim.
- Pitch to select attitude, trim to relieve pressure.
- Any adjustment to pitch or power should be followed with an adjustment to trim.

Engine Handling

- Highlight the red line on the RPM gauge.
- Ensure that the student understands the negative effects of "redlining" the engine.
- When stationary on the ground, RPM should be set to 1000 RPM.
- Do not lean the mixture below 3,000 ft.
  - For leaning guidance, refer to the PIM and the leaning guidance on the FIT hub.

Effect of Power and Airspeed

- For an increase in power.
  - The pitch will increase and there will be a yaw to the left.
  - To correct you need forward pressure on the yoke and right rudder.
- For a decrease in power.
  - The pitch will decrease and there will be a yaw to the right.
  - To correct you need back pressure on the yoke and left rudder.
- For an increase in airspeed.
  - The pitch will increase and there will be a yaw to the right.
  - To correct you need forward pressure on the yoke and left rudder.
- For a decrease in airspeed.
  - The pitch will decrease and there will be a yaw to the left.

- To correct you need back pressure on the yoke and right rudder.

Effect of Flaps

Flap Setting / Effects	Lift	Nose	Elevator (initially)	Drag	Airspeed	Elevator (due to drag)
0 - 10 Degrees	↑↑↑	↑↑	Forward	↑	↓	Backward
10 - 25 Degrees	↑↑	↑↑	Forward	↑↑	↓↓	Backward
25 - 40 Degrees	↑	↑	Forward	↑↑↑	↓↓↓	Backward
40 - 25 Degrees	↓	↓	Backward	↓↓↓	↑↑↑	Forward
25 - 10 Degrees	↓↓	↓↓	Backward	↓↓	↑↑	Forward
10 - 0 Degrees	↓↓↓	↓↓	Backward	↓	↑	Forward

Operation of Mixture Control

- The mixture lever has a red knob and is located to the right of the throttle lever.
- Moving the mixture lever forward results in a rich fuel/air mixture, while moving the mixture lever backward results in a lean fuel/air mixture.
- <u>Emphasize that we do not lean the mixture below 3000 feet.</u>

Operation of Carburetor Heat

- The carburetor heat lever is located to the right of the mixture lever.
- To operate, pull the lever downward to the on position or upward to the off position.
- <u>Review when carburetor heat is required (show in checklists).</u>

Operation of Environmental Controls

- Demonstrate how to operate the cabin heat and defroster.

Other Controls as Applicable

- Adjusting audio panel power selection, volume, squelch, Garmin430 basic usage (com selection, frequency changes, power selection).

<u>Post-Brief</u>

- Student self-critique.
- Instructor critique.
- Questions from the student.
- Lesson grading.
- Logbook completion.
- Schedule the next lesson.
- Preview the next lesson → 105.
- Assign homework:
    - Review primary and secondary flight controls.

  - Have the student memorize the passenger briefing, engine fire during start briefing, and the takeoff briefing.
  - Review chapter 03 of the Airplane Flying Handbook - Basic Flight Maneuvers:
    - The four fundamentals.
    - Effects and use of flight controls.
    - Attitude flying.
    - Straight and level flight.

Completion Standards

This lesson is complete when the student is able to:

1. Obtain a weather briefing by utilizing 1-800-WX-BRIEF and complete a TOLD card with minimal instructor guidance.
2. Describe the elements of aircraft performance charts, weight and balance and environmental factors that affect performance.
3. Demonstrate an understanding of checklist procedures and effects of primary, secondary, and ancillary flight controls with instructor guidance.